VERIFIED BIBLIOGRAPHIC & EMPIRICAL EXTRACT (Keith_2018_Joule_Liquid_DAC)
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Title: A Process for Capturing CO2 from the Atmosphere
Authors: David W. Keith, Geoffrey Holmes, David St. Angelo, Kenton Heidel (2018)
Source: Joule, Volume 2, Issue 8, Pages 1573–1594
DOI/URL: https://doi.org/10.1016/j.joule.2018.05.006
Sample: N = 1 Mt-CO2/year commercial direct air capture facility

EXTRACTED EMPIRICAL METRICS & TABLE COORDINATES:

• Process Architecture: Aqueous potassium hydroxide (KOH) liquid sorbent coupled to a calcium caustic recovery loop (pellet reactor, oxy-fired calciner, slaker).
• Chemical Unit Reactions: Air contactor: CO2 + 2KOH -> K2CO3 + H2O; Pellet reactor: K2CO3 + Ca(OH)2 -> 2KOH + CaCO3; Oxy-calciner: CaCO3 -> CaO + CO2 at 900 deg C; Slaker: CaO + H2O -> Ca(OH)2.
• Thermal & Electrical Duty: Delivered at 15 MPa pipeline pressure: Baseline all-natural gas configuration requires 8.81 GJ natural gas / t-CO2 captured. Hybrid configuration requires 5.25 GJ natural gas + 366 kWh electricity / t-CO2 captured (or 360-550 kWhe/t).
• Combustion Co-Capture: Oxy-fired CFB calciner co-captures ~0.48 t-CO2 from on-site natural gas fuel alongside each 1.0 t-CO2 captured from the ambient atmosphere (total delivered ~1.48 t-CO2/t-air CO2).
• Solid Sorbent TVSA Comparison: Deutz & Bardow (2021, Nature Energy, doi:10.1038/s41560-020-00771-9) evaluate solid amine TVSA (Climeworks) requiring 5.4-9.0 GJth (1.5-2.5 MWhth) low-temperature heat (80-110 deg C) and 450-750 kWhe/t CO2 for fans/vacuums.
• Life Cycle Removal Efficiency: Net removal efficiency reaches 93.1% under geothermal power (Hellisheidi) and 85.4% under waste heat (Hinwil). Under average grid power (~500 g CO2/kWh), net efficiency falls below 40% or becomes net positive.

VERBATIM SOURCE QUOTE / TABLE RECORD:

We describe a process for capturing CO2 from the atmosphere in an industrial plant. The design captures ~1 Mt-CO2/year in a continuous process using an aqueous KOH sorbent coupled to a calcium caustic recovery loop. We describe the design rationale, summarize performance of the major unit operations, and provide a capital cost breakdown developed with an independent consulting engineering firm. We report results from a pilot plant that provides data on performance of the major unit operations. We summarize the energy and material balance computed using an Aspen process simulation. When CO2 is delivered at 15 MPa, the design requires either 8.81 GJ of natural gas, or 5.25 GJ of gas and 366 kWhr of electricity, per ton of CO2 captured. Depending on financial assumptions, energy costs, and the specific choice of inputs and outputs, the levelized cost per ton CO2 captured from the atmosphere ranges from 94 to 232 $/t-CO2.